

CH3 Taylor Expansion of Analytic Functions

3.1 Theorem If

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n \quad (z \in B(a, R)) \quad (1)$$

and if $0 < r < R$, then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(a + re^{i\theta})|^2 d\theta. \quad (2)$$

Proof1: We have

$$f(a + re^{i\theta}) = \sum_{n=0}^{\infty} c_n r^n e^{in\theta}. \quad (3)$$

For $r < R$, the series (3) converges uniformly on $[-\pi, \pi]$. Hence

$$c_n r^n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(a + re^{i\theta}) e^{-in\theta} d\theta \quad (n = 0, 1, 2, \dots),$$

and (2) is seen to be a special case of Parseval's formula(real analysis edition).

□

Proof2: Without loss of generality, let $a = 0$. We have

$$\frac{1}{2\pi} \int_0^{2\pi} z^n \bar{z}^m d\theta = r^{n+m} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 0, & m \neq n, \\ r^{2n}, & m = n. \end{cases}$$

Since $f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n$ is absolutely and uniformly convergent on $B(0, R)$, the order of summation and integration can be interchanged. Therefore,

$$\begin{aligned}\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta &= \frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{n=0}^{\infty} a_n z^n \right) \cdot \left(\sum_{m=0}^{\infty} \overline{a_m} \bar{z}^m \right) d\theta \\ &= \frac{1}{2\pi} \sum_{m,n=0}^{\infty} a_n \overline{a_m} \int_0^{2\pi} z^n \bar{z}^m d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.\end{aligned}$$

□

Corollary If $f(z) = \sum_{n=0}^{\infty} c_n z^n$ is a bounded holomorphic function on $B(0, 1)$, then $\sum_{n=0}^{\infty} |c_n|^2 < \infty$.

Proof: Suppose

$$|f(z)| \leq M, \forall z \in B(0, 1).$$

From above , we have

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} M^2 d\theta = M^2, \forall r \in (0, 1).$$

By the series version of **Levi's Theorem**, it follows that

$$M^2 \geq \lim_{r \rightarrow 1^-} \sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \sum_{n=0}^{\infty} |c_n|^2.$$

□